

HOMEWORK ASSIGNMENT 4 – Solutions

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

Instructions:

- Kindly recall that homework assignments for this course are **not turned in**, and **not graded**.
 - However, problems from homework sets will form the basis of problems that appear on the weekly quizzes; as such, it is in your best interest to complete the homework sets on your own
 - Solutions to this assignment will be released in the morning of Friday September 18.
 - Kindly note that there will be **no quiz** on Monday September 21. However, the material this homework tests you on could absolutely show up on the **Midterm** on Wednesday September 23.
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1. **Harley's Return.** In this problem, we'll revisit the study from Lecture Activity 5 ("Can Dogs Understand Human Cues?") and reformulate it in the language of hypothesis testing. Recall the setup: Harley was presented with two cups, one of which the experimenter bowed towards. The research question was: can Harley understand human cues? The data was: Harley chose the "correct" cup (i.e. the cup the experimenter bowed toward) in 9 of the 10 total trials.

- (a) What proportion of Harley's trials resulted in him selecting the correct cup (what we are labeling as a "success")? Report the appropriate value, as well as its proper mathematical symbol.

Solution: $9/10 = 0.9 = 90\%$ of Harley's trials resulted in a "success," which we are defining to be "him selecting the correct cup." This value is an observed value of $\hat{p}$, the sample proportion.

- (b) What is the parameter of interest? Provide a verbal description in the context of the study, and also the proper mathematical symbol.

Solution: The parameter of interest is **the long-run proportion of times Harley picks the correct cup, under repeated trials**. We use the symbol π to denote this; if we wanted to be more explicitly, we could use a subscript like π_{correct} .

- (c) Write down the null and alternative hypotheses using the parameter you defined in part (b) above.

Solution:

- The null hypothesis (H_0) is that "Harley is just guessing;" i.e. that $\pi_{\text{correct}} = 0.5$. So, for the null distribution, we would write

$$H_0 : \pi_{\text{correct}} = 0.25$$

- The alternative hypothesis (H_A) is that "Harley is doing something more than just guessing;" i.e. that $\pi_{\text{correct}} > 0.5$. So, for the alternative distribution, we would write

$$H_A : \pi_{\text{correct}} > 0.25$$

For Parts (c) onwards: Return to the One Proportion Applet, and conduct 10,000 repetitions of the chance model to obtain a simulated null distribution. Make sure that you have the option "Proportion of heads" selected (and not "Number of heads"). **Don't close out of your plot yet** - you'll be asked to modify various parts of it in the upcoming parts of this question!

(d) What value did you use for the probability of success (π)?

Solution: We conduct our simulation “under the null;” i.e. assuming that the null distribution is correct. Therefore, we should set the probability of success in our simulation to be **0.5**.

(e) What value did you use for the sample size (n)?

Solution: The sample size is **10**, representing the 10 trials Harley conducted.

(f) What does a dot on the simulated null distribution represent? Answer using the context of the study.

Solution: A dot represents the proportion of times (out of 10) Harley would pick the correct cup.

Note: it’s important that you include the sample size in your description, as otherwise it is incomplete.

(g) When trying to use the applet to approximate the p –value, should you use the $\leq$, $=$, or $\geq$ option in the applet? Explain.

Solution: The direction of extremity matches the direction of the alternative hypothesis, meaning we should use the $\geq$ option in the applet.

(h) Calculate your approximate p –value.

Solution: Answers may vary. Here is a screenshot of my work:

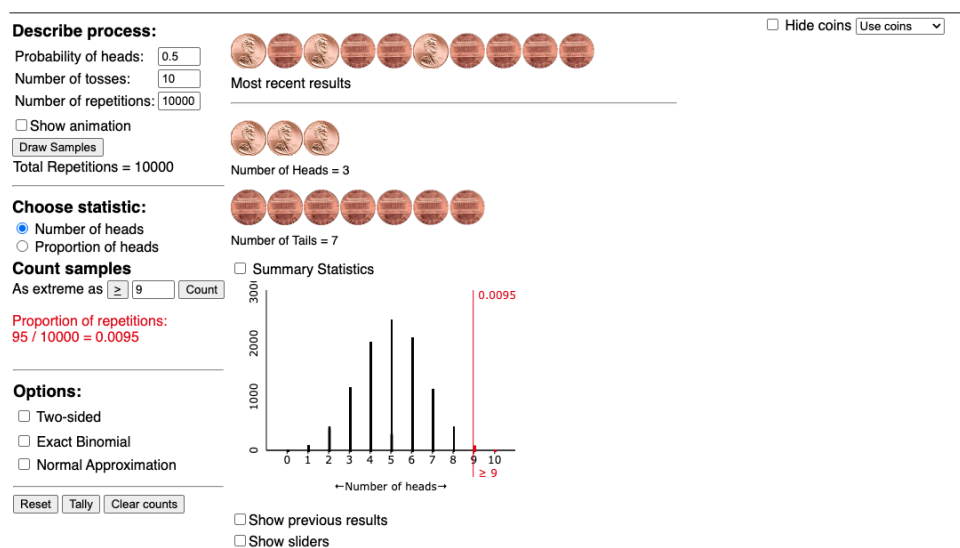


Figure 1: Screenshot of the correct inputs and output.

Because the proportion of the 10,000 simulated $\hat{p}$ values that were greater than or equal to 0.1 was 0.0095, this represents the approximate p -value.

(i) According to the guidelines posted in Lecture Activity 6 (and in the textbook), what is the strength of evidence?

Solution: An approximate p -value of 0.0095 provides **very strong evidence against the null**.

- (j) Summarize the conclusion that you draw from this study based on your analysis. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) approximate p -value (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p -value, (4) reject/fail to reject the null hypothesis, and (5) overall conclusion in context. Make sure you refer to the **parameter** in your answer.

Solution: The sample proportion of times Harley picked the correct cup (out of 10 trials) was 0.9. Based on this value, we obtained a very small approximate p -value of around 0.0095. This provides very strong evidence to reject the null hypothesis, and state that the long-run proportion of times Harley would pick the correct cup is greater than 0.5, for trials conducted under similar conditions to those described in the original study. In other words, Harley appears to be doing something more than just guessing each time.

2. **Right-Handedness.** According to a somewhat old 2009 article (The Left Brain Knows What the Right Hand is Doing), 90% of people are right-handed. We'll take this to mean the authors claim that 90% of all people in the world are right-handed. Suppose we want to test this article's claim against the alternative that the true proportion of right-handed individuals in the world is less than 0.9. For our data, we'll consider the Welcome Survey I sent out earlier this semester to my three sections (S17, S18, and S19) of STAT 1110, in which 69 of the 84 respondents indicated they are right-handed.

- (a) Describe one observational unit in the context of this study.

Solution: One observational unit is **one person**.

- (b) Describe the population of interest, in the context of this study.

Solution: The population is the set of **all people in the world**.

- (c) Describe the sampling frame, in the context of this study.

Solution: The sampling frame is the **set of all STAT 1110 students enrolled in one of my (Dr. Ethan's) sections**.

- (d) Define the parameter of interest. Be sure to use proper mathematical notation.

Solution: The parameter of interest is π_{RH} , the proportion of all people in the world who are right-handed.

- (e) Write down the null and alternative hypotheses in terms of the parameter you defined in part (d) above.

Solution: The null hypothesis is $H_0: \pi = 0.9$ (corresponding to the article's claim) and the alternative hypothesis is $H_a: \pi < 0.9$ (corresponding to the claim that fewer than 90% in the world are right-handed).

Let's start off with a simulation-based approach. Return to the One Proportion Applet, and conduct 10,000 repetitions of the chance model to obtain a simulated null distribution. Make sure that you have the option "Proportion of heads" selected (and not "Number of heads"). **Don't close out of your plot yet** - you'll be asked to modify various parts of it in the upcoming parts of this question!

(f) Calculate the approximate p -value.

Solution: Answers may vary. Here is a screenshot of my work:

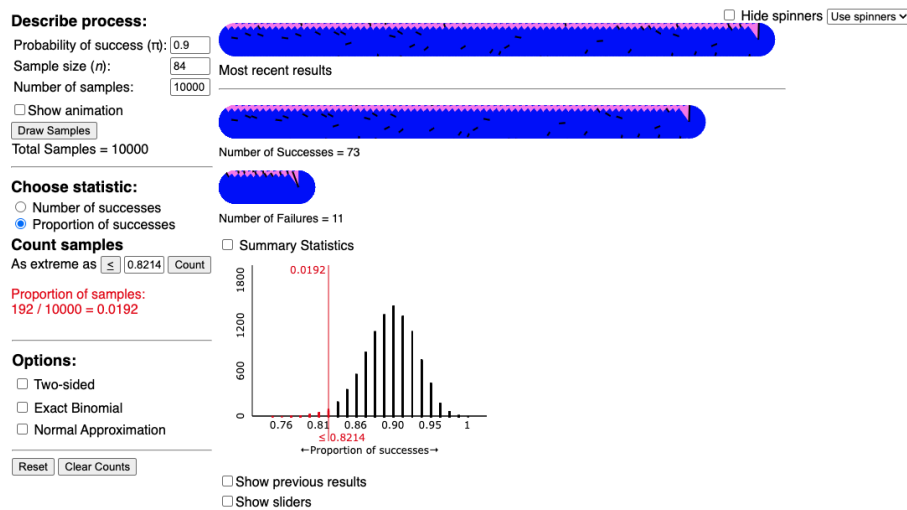


Figure 2: Screenshot of the correct inputs and output.

So, my approximate p -value is **0.0192**

(g) Mark the correct options on the applet (it's up to you to remember what these are, from Lecture Activity 7) to calculate the mean and standard deviation of the 10,000 simulated $\hat{p}$ values.

Solution: We check the "Summary Statistics" box option:

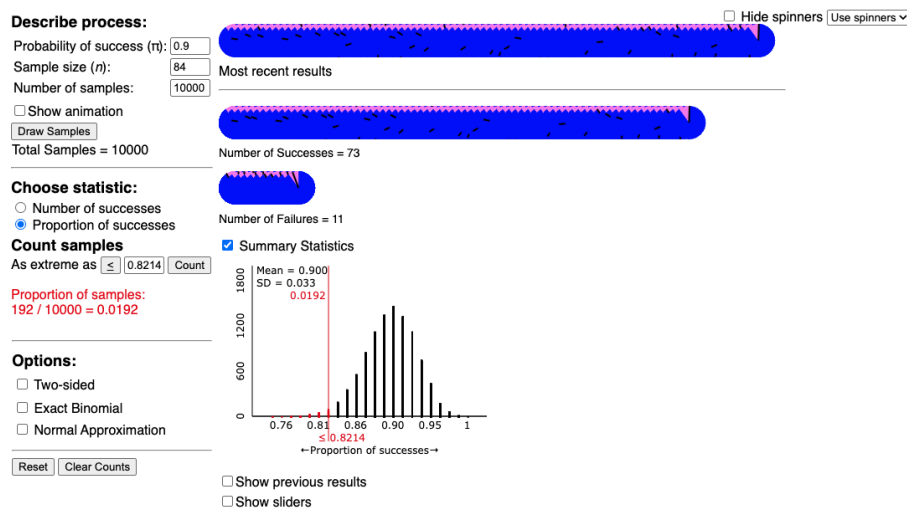


Figure 3: Screenshot of the correct inputs and output.

So, the mean is **0.900** and the standard deviation is around **0.033**

In the remaining parts of this problem, we'll carry out a theory-based hypothesis test of the hypotheses posited in part (e) above.

- (h) Verify that the validity conditions are satisfied.

Solution: We require at least 10 successes and at least 10 failures. The number of successes in our observed sample is 69, which is at least 10; the number of failures is $(84 - 69) = 15$, which is also at least 10. Therefore, the validity conditions are satisfied.

- (i) What are the center (mean) and standard deviation of the true/theoretical null distribution, and how can you tell?

Solution: If the validity conditions are met, then the true/theoretical null distribution is centered at π_0 with a standard deviation of $\sqrt{\pi_0(1 - \pi_0)/n}$, where π_0 is the null value. Since the null value is $\pi_0 = 0.9$, this means the true/theoretical null distribution is centered at 0.9 with a standard deviation of $\sqrt{(0.9) \cdot (0.1) / 84} \approx 0.03273268$

- (j) How do your answers to parts (i) and (g) compare? Would they necessarily be *exactly* the same, or would we merely expect them to be *quite close* to each other?

Solution: Since the validity conditions are met, we would expect our answers to be very close. However, we wouldn't be surprised if they are slightly different because of sampling variability.

- (k) Calculate the value of the standardized score.

Solution:

$$\begin{aligned} z &= \frac{(\text{statistic}) - (\text{mean of null distribution})}{(\text{standard deviation of null distribution})} \\ &= \frac{\frac{69}{84} - 0.9}{0.03273268} = \frac{-0.07857143}{0.03273268} \approx \mathbf{-2.40} \end{aligned}$$

- (l) Provide an interpretation of the standardized score you just calculated. **Hint:** think about how standardized scores represent the number of ____'s an observed statistic value is away from the ____.

Solution: A standardized score represents how many standard deviations an observed statistic value is away from the mean, all under the null.

- (m) According to the conventions established in Section 1.3 of the textbook, would we classify the observed value of the statistic as being in the tails of the null distribution? Explain briefly.

Solution: The convention we're using is that observations that fall more than 2 or 3 standard deviations from the mean can be considered in the tail of the distribution. Since our observed sample proportion is around 2.4 standard deviations to the left of the mean of the null distribution, we can classify it as being in the tails.

- (n) Based on your answer to the previous part, do you think there is sufficient evidence against the null to reject it in favor of the alternative? Why?

Solution: Yes; observations that fall in the tail of the null distribution are unlikely to occur under the null. As such, since our observed statistic was pretty unlikely under the null, we have evidence against the null.

- (o) Navigate to the Theory-Based Inference Applet, and fill in the appropriate values. Click “calculate” and report the true/exact p -value. To show your work, include a screenshot of your entire screen showing the inputs and outputs.

Solution:

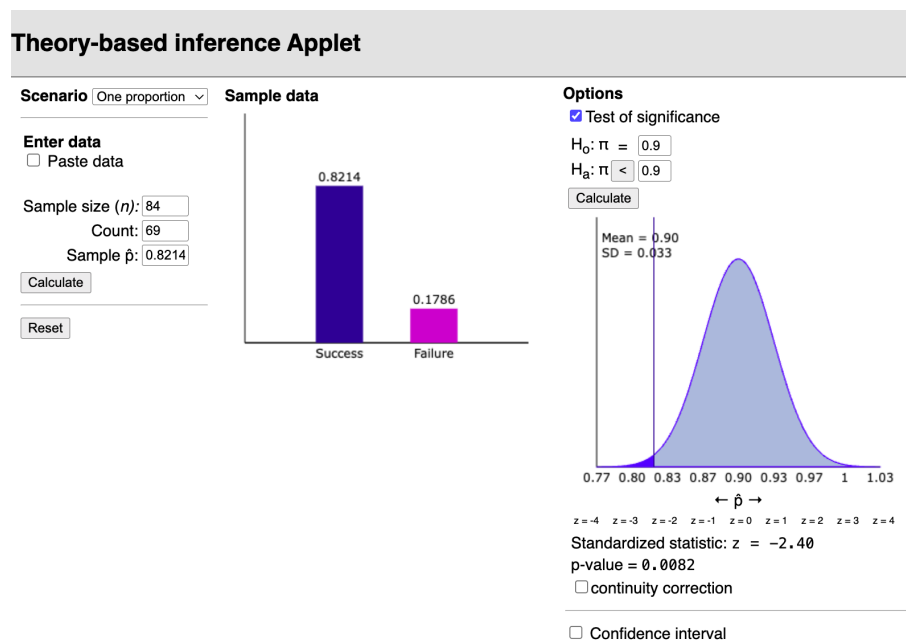


Figure 4: Screenshot of the correct inputs and output.

The p -value is **0.0082**.